

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could coordinated cortical activity support one internally represented situation? We distinguish coherence, agreement between local contents, and accurate self-reconstruction, and develop conditional connections among them. Declared actuator modes give a necessary installed-energy condition for a coherent stationary branch. Phase encoders shared on overlaps bound content disagreement; compatible spatial measures form one global state. A contracting reconstruction that fixes this state makes it the unique fixed point, while accuracy across a state family requires distinguishable encodings. Lean 4 checks the results and their composition under explicit connecting hypotheses. Numerical controls delimit their observational use: finite ramps delay coherence, distinct mechanisms share stationary phase distributions, a winding field and a uniform one are stationary at one coupling with opposite resultants, and content agreement requires independently validated decoders. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. An extracellular-field hypothesis links awakening to increasing coupling. Its test requires calibrated geometry and phase dynamics, together

with measurements of content agreement, self-encoding and experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are

21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 8.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand's relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator's stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable

44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
51 composes them, stating the thermodynamic constraints and all eight con-
52 necting hypotheses per branch; its supporting models of feedback, learning
53 and resource supply have distinct state spaces and assumptions. Section 6
54 then examines what the observables can establish, before Section 7 devel-
55 ops the cortical hypothesis and an awakening experiment. The hypothesis is
56 that extracellular contraction increases field-mediated coupling; its direction,
57 rate and relation to recovery require independent calibration. Supplemen-
58 tal Material organizes the constructions and controls behind the claims they
59 support.

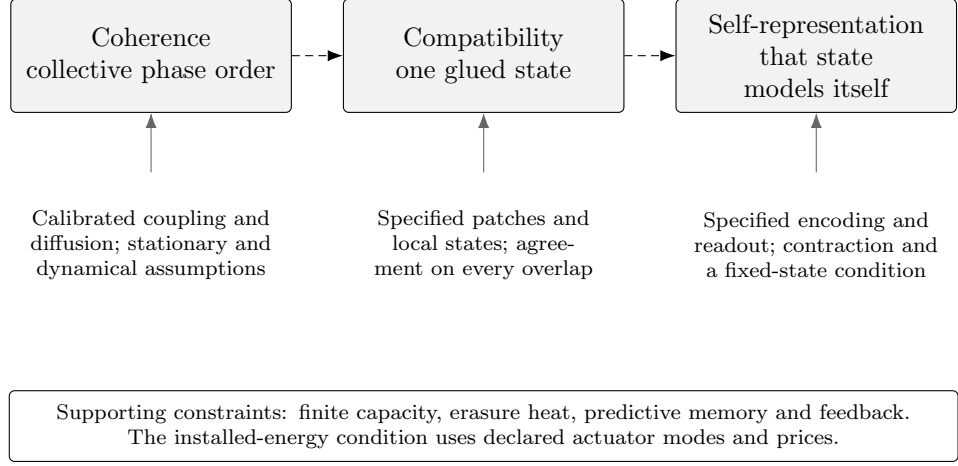


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

2 Coherence: physical coupling and its resource condition

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency. At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero.

83 These linear estimates concern the limiting equation. Relating them to finite
84 stochastic trajectories requires a dynamical mean-field limit and branch selec-
85 tion; exchange symmetry alone does not establish the needed independence
86 (Sznitman, 1991).

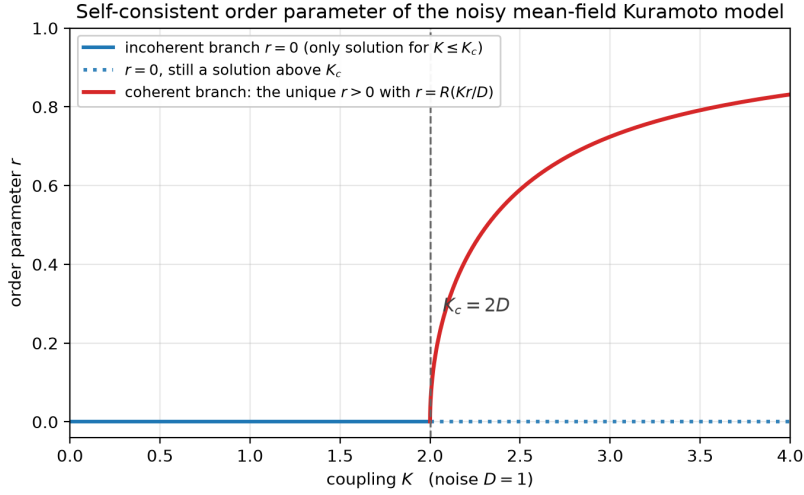


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

87 2.2 Installed energy bounds mean coupling

88 A spatial model replaces the finite sum by integration against a kernel. One
89 declared implementation carries response modes with continuous profiles ϕ_i ,

90 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

91 The same configuration sets the reciprocal kernel and stored energy. An
 92 executed transition changes the kernel through its changed occupancies and
 93 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 94 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 95 pair reconstruction on refining spatial partitions converges to its product-
 96 measure integral under the specified compactness and sampling assumptions;
 97 the refinement index is independent of the number of control updates.

98 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 99 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 100 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

101 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 102 coherent stationary branch and any growing Fourier mode about the uniform
 103 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 104 is necessary: energy above it need not produce sufficient coupling because
 105 the bound can be loose.

106 The quantity constrained is installed energy. Dissipated heat does not
107 determine it, and κ is supplied by the mode profiles, prices and substrate
108 normalization. Applying the scalar threshold to heterogeneous or signed spa-
109 tial dynamics requires a justified reduction. Identifying the installed kernel
110 with a cortical field is a separate physical commitment.

111 **3 Compatibility: from coordinated activity** 112 **to a global description**

113 **3.1 Why common timing is insufficient**

114 For overlapping patches U and V , compatibility means that restricting either
115 local description to $U \cap V$ gives the same object. Descriptions can agree on
116 shared quantities while contributing different information elsewhere. Com-
117 patible sections of the specified sheaf have a unique global extension. On
118 a finite cover, local finite spatial measures agreeing as measures on every
119 overlap also have a unique finite-measure extension whose sheaf image is the
120 glued section. No mass-one normalization is imposed.

121 In the cup example, visual and reaching descriptions must express a
122 shared quantity, such as cup location, in common coordinates. Restriction
123 retains that quantity while dropping details particular to each description.
124 Agreement permits surface appearance and hand configuration to contribute
125 to one situation. The mathematical counterexample assigns the same phase

126 to every patch while two local mass profiles disagree on a shared site. Those
 127 profiles have no common extension. Because they are chosen independently
 128 of phase, synchrony places no constraint on their disagreement.

129 This distinction limits the biological interpretation. Receptive territories
 130 and content domains must be related to spatial patches, and the descriptions
 131 must support the required extension. Matching overlap marginals of de-
 132 coded probability laws need not determine a unique global joint distribution
 133 (Abramsky & Brandenburger, 2011).

134 **3.2 When coherence does constrain content**

135 A declared relation between phase and content gives a positive result. Let
 136 contents take values in a real normed vector space E . Patch i reads an
 137 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 138 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 139 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 140 sites; their equality on overlaps is a supplied relation between descriptions of
 141 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 142 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 143 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

144 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 145 agree exactly. The factor N makes this estimate potentially weak in large
 146 populations; it supplies a pointwise guarantee from a population average.
 147 The population in it need not be the whole array. A patch carries an order pa-
 148 rameter of its own, and the same bound read there carries that patch's resul-
 149 tant and that patch's site count (`chord_le_of_patch_coherence, compatible_of_patch_coherence`
 150 so a large population observed through small patches is constrained by the
 151 small count. A shared scalar encoder is a special case. Dropping the overlap
 152 relation permits disagreement even at perfect locking. A common nonex-
 153 pansive linear projection preserves agreement, but agreement of a projection
 154 does not establish agreement of the full vector.

155 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 156 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 157 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 158 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

159 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 160 duce an initial mismatch. Approximate observations leave a nondecaying
 161 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 162 update to the phase process.

163 3.3 What approximate agreement determines

164 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
165 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
166 Weights that vanish outside their patches and sum to one at each site select
167 a global profile by weighted averaging. The selection lies within ε of each
168 patch there, and any two such selections differ by at most ε , independently
169 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
170 $L\varepsilon$ across selections.

171 At zero tolerance, the selection is the unique exact extension. At every
172 positive tolerance there are families with no exact extension; selection then
173 gives an approximation whose dependence on weights is bounded. Supple-
174 mental Material gives a three-site witness attaining the bound. The mass-
175 profile construction uses the scalar case of the content dynamics and addi-
176 tionally requires nonnegativity of the initial profiles and observations. The
177 general vector result supplies no identification with the spatial-measure sheaf.

178 These constructions concern declared objects: a specified cover, contents
179 valued in a normed space, and encoders related on overlaps. They select
180 neither the neural cover nor the decoder. Identifying their contents with
181 perceptual content, and establishing that the identification bears on experi-
182 ence, is a further modelling task; Section 6 sets out what a measurement of
183 it would have to establish.

184 4 Self-representation: a state that models it- 185 self

186 4.1 The reflexive construction

187 Compatibility gives a global state. Reflexivity relates it to an encoding within
188 that state. Equip global sections with a complete metric. If the additional
189 restriction-resonance condition identifies encoding with restriction to a region
190 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

191 The region and readout are fixed across the state space. If F contracts and
192 fixes the section s_* formed from the local descriptions, Banach's theorem
193 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
194 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
195 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
196 one for $\tau > 0$. Constructing a map with that factor remains a modelling
197 obligation.

198 In the cup example, the global state includes the represented scene and the
199 perceiver's relation to it. A local encoding must retain enough information
200 for the readout to recover that modelled state; cup position alone would be
201 insufficient if hand configuration and viewpoint could not be reconstructed.

202 For the same map with Lipschitz factor $q < 1$, approximate reconstruction

203 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1-q}. \quad (9)$$

204 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 205 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 206 weakens.

207 4.2 Reconstruction across a family of states

208 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 209 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 210 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

211 Thus M relevant states pairwise more than 2ε apart require at least M
 212 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 213 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 214 existing self-prediction map.

215 This is an additional accuracy criterion on a family, with no requirement
 216 for multiple exact fixed points. A blind encoder or constant readout fails it
 217 for separated states despite having a unique fixed point.

218 4.3 Agreement after a change takes communication

219 Snapshot compatibility does not specify how quickly descriptions respond
220 when a shared quantity changes. In a synchronous network whose updates
221 read only the site’s own state and declared neighbours’ messages, two initial
222 configurations agreeing on a site’s causal past through T rounds produce the
223 same state there at round T . If two fresh interventions outside that past
224 require reports separated by more than 2ε , no report from that site can be
225 within ε of both. This follows from the reconstruction bound in Section 4.2.

226 A three-site delay line witnesses failure after one round and exact report-
227 ing after two; cutting the path prevents delivery at every round. A report
228 may still happen to be correct about one fixed world. The bound concerns
229 responsiveness across interventions.

230 Throughout this section *prediction* names a reconstruction map; reading
231 it as temporal neural prediction would require dynamics. The relevant family
232 of states, the metric, the tolerance, the communication graph and the dead-
233 line are all declared inputs, as are the represented variables and the fidelity
234 counted as accurate. Reading the resulting code count as physical memory
235 capacity requires a hardware model, and a readout that varies with the state
236 requires a further one.

237 5 Thermodynamics and composition

238 The phase and representation results concern coupling, content disagreement
239 and reconstruction error. Thermodynamics constrains the physical opera-
240 tions that could implement them. Finite-register capacity obeys $H(p) \leq$
241 $\log |X|$; erasure heat depends on specified system–bath assumptions (Lan-
242 dauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result con-
243 cerns a field reaching disconnected vacuum components (Kibble, 1976). The
244 composition connects these objects through the explicit hypotheses in Ta-
245 ble 1.

246 5.1 Feedback changes the information constraint

247 For passive memory X_t driven by an environmental signal S_t , the predictive
248 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

249 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
250 et al., 2012). Zero retained information also satisfies it, so a task and a
251 learning rule require additional specification. Relating this bound to Lan-
252 dauer heat requires a common thermal scale and budget, and memory waste
253 bounded by erased entropy.

254 An acting agent changes the signal it will receive. Specify policy, world,

255 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{12}$$

256 The second conditional term vanishes in the passive model. With feedback,
257 the signed difference can be negative as action creates correlations with the
258 future. A causal reading requires an environmental state sufficient to describe
259 the relevant dynamics.

260 For a controlled substep, the active thermodynamic bound instead uses
261 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
262 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
263 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
264 2013). Every forward path of positive probability must have a positive-
265 probability reverse for this finite identity. A missing reverse gives infinite
266 divergence. The supplement develops the support conditions and composes
267 actuation with memory updates on their shared intermediate law.

268 **5.2 Learning and repeated operation need explicit re-** 269 **sources**

270 The finite examples distinguish selecting a feasible policy, relaxing a register
271 toward an encoded reward, and learning an unknown task from observa-
272 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

A phase-derived sensor makes the informational distinction concrete: an informative readout and a blind control have the same phase order but different learning performance. Attaching the learner’s register to the actuator of Section 2.2 ties reward, installed kernel and work to its own executed law. The learning witness rises above the necessary installed-energy bound while its matched control remains on the exclusion line. These finite constructions, detailed in Supplemental Material, leave the evolving feedback between phase dynamics, installation and sensing to be specified.

For that model, common energy E , thermal scale θ , local detailed balance and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles,

while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply automatically.

5.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.

Table 1 (continued)

Connection	Status	Required assumption
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

307 lap compatibility and a reflexive boundary specifying encoding and readout.
 308 They conclude **UnifiedSelf**: the glued section is the unique fixed point of
 309 self-prediction. Compatibility and accuracy at that section enter as premises;
 310 uniqueness follows. Identifying encoding with restriction needs the separate
 311 resonance condition of Eq. (8).

312 Each of the eight edges is supplied rather than derived from the one above
 313 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
 314 pendent roots rather than a step from one to the other; both E45 discharges
 315 take their convergence from refinement data and not from the thermody-
 316 namic bound; E78 states that coherence relaxes the supplied cover, and the
 317 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
 318 coders already shared on overlaps. The inferential content of the composition
 319 is its last step, where contraction and a fixed glued section give uniqueness.
 320 What composing them establishes is that the premises are consistent, named
 321 in one place and machine-checked together, and not that coherence follows
 322 from installed energy or self-representation from coherence.

323 Joint witnesses combine a register, boundary, refining mesh and small
 324 spatial cover; active witnesses add controlled thermal updates and a priced
 325 installation. They establish satisfiability under their stated identifications.
 326 Mesh convergence follows from refinement data, and the witnesses establish
 327 no common cortical mechanism. Supplemental Material maps the claims to
 328 their Lean declarations and records the counterexamples. The development
 329 declares no additional axioms beyond its libraries' foundational axioms.

330 6 What the observations establish

331 The composition holds under its premises. Whether observation can establish
332 those premises is a separate question, and the three requirements it connects
333 concern different objects: phase dynamics, local descriptions and internal
334 encodings. Their observables must retain these distinctions. The following
335 controls show which mechanisms remain indistinguishable, which temporal
336 approximations fail, which states one observable cannot separate, and what
337 independent content measurements must establish.

338 6.1 Stationary distributions leave the mechanism un- 339 determined

340 Equation (2) gives a concentration–coherence relation with no fitted shape
341 parameter. Estimate r from circular resultant length and a from density
342 shape: inverting R from the measured r would make agreement automatic.
343 Even independently estimated agreement identifies a statistical family shared
344 by different mechanisms.

345 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
346 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
347 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
348 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
349 and diffusion D_0 have that same density. Their stationary distributions co-
350 incide window by window despite different causes.

Resolved dynamics supply additional information. In the ideal model with continuous paths, quadratic variation accumulates at rate $2D$ and conditional increment means constrain drift. Separating endogenous coupling from a matched common drive additionally requires a measured drive or controlled perturbation; identifying electromagnetic coupling requires evidence specific to that interaction. Supplemental Material gives the equivalence construction and its numerical check.

6.2 Finite ramps change the observable onset

The square-root law in Eq. (3) describes stationary solutions. A changing coupling requires time for the phase distribution to approach those solutions. In a deterministic mean-field control that crosses the threshold symmetrically, none of the scanned rates meets the declared tracking tolerance: slowing the crossing does not remove the terminal residual under that initialization. A slow control variable therefore supplies no general guarantee of branch tracking at onset. Supplemental Material gives the residual, rate range and admissible rates away from threshold.

The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas at four coupling speeds. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32 replicas escape within the fastest window. The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents differ from those of the exact stationary branch evaluated on the same grids.

373 At the fastest speed the ensemble mean never reaches the fit window, so its
374 fitted slope does not estimate stationary onset.

375 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
376 makes the apparent exponent depend on ramp speed and fitting window.
377 The specified deviation-based critical ramp speed is not bracketed by this
378 sweep. With fluctuating coupling, the relevant stationary comparison also
379 depends on rate: the mean coupling approximates order in the tested fast
380 regime, while averaging stationary order over coupling works for the tested
381 slow drives confined to one side of threshold. Slow crossing drives can satisfy
382 neither approximation. These are finite-horizon observations, with controls
383 and parameters in Supplemental Material.

384 Additional controls address integration bias, population size and spatial
385 or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
386 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
387 rameters and controls in Supplemental Material; they supply no cortical cal-
388 ibration.

389 **6.3 A winding state leaves the order parameter am-** 390 **biguous**

391 The order parameter r is indifferent to where a phase sits in space. A field
392 winding once across a periodic sheet therefore has $r = 0$ while every patch
393 of it stays locked. Such a field is not an artificial construction here. On

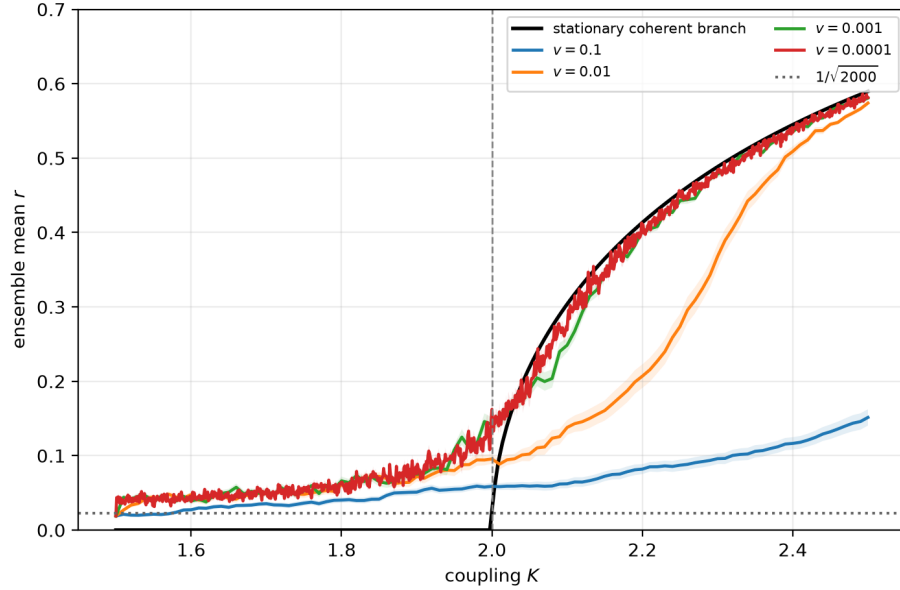


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

394 a torus a twist is stationary for any isotropic symmetric kernel at identical
395 frequencies: the coupling term is odd in the separation against a kernel even
396 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
397 this for any finite periodic array, this sheet included. A twist satisfies the
398 positivity and identical-frequency hypotheses. It violates only the quarter-
399 turn initial spread that the convergence result additionally requires before
400 placing the limit at the uniform state.

401 On the sheet of Section 2's spatial control, a twist survives below an in-
402 terpolated decay length of 0.1871 mm and unwinds above it. That boundary

403 is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the same
 404 coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start and
 405 $r = 0.0083$ from a twisted one. Both are stationary, with patch-local order
 406 0.9015 and no winding defect. At fixed K the observable therefore takes two
 407 values spanning nearly its whole range. Coupling does not determine the
 408 resultant, and a low resultant does not by itself distinguish weak coupling
 409 from a winding field.

410 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
 411 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 412 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realisations.
 413 Heterogeneity shortens the range that sustains a wave without removing it.

414 Which windings persist has a criterion behind it. Linearising the phase
 415 model about a winding returns a coupling of the same translation-invariant
 416 form, each separation reweighted by the cosine of the phase the winding
 417 advances across it (`hasDerivAt_charJacobian`). That reweighted ker-
 418 nel is again even in the separation, so the same characters diagonalise
 419 it and every mode’s rate is one of its eigenvalues. Where the wind-
 420 ing advances by at most a quarter turn across every separation the cou-
 421 pling reaches, the reweighted kernel is nonnegative and no rate is positive
 422 (`charJacobian_stable_of_quarter_turn`); where some rate is positive, that
 423 mode grows without bound (`charJacobian_growing_mode`). Moving cou-
 424 pling mass to longer separations is what carries a phase advance past a
 425 quarter turn, and that is the mechanism the decay-length sweep varies. The

426 criterion bounds rates rather than trajectories: asymptotic stability is not
 427 proved, so the states that persist occupy a narrower range than it admits,
 428 and the boundary quoted above is measured and not derived from it.

429 The same reweighted kernel measures the potential's curvature. A wind-
 430 ing has $r = 0$ where a global minimiser of the Lyapunov potential has $r = 1$,
 431 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 432 tial data does not merely decline to cover a winding, it must exclude it. Along
 433 a mode, the potential's second variation is that kernel's quadratic form, so a
 434 growing mode is a direction of falling potential and a winding carrying one
 435 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

436 An amplitude layer decides which of these states exist, and leaves what
 437 the resultant reports unchanged. A phase state carries no amplitude. Every
 438 oscillator sits on the unit circle, so a field with an amplitude node has no rep-
 439 resentation in these dynamics. The real Stuart–Landau extension supplies
 440 one on the same coupling, and leaves intact the gradient structure the con-
 441 vergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\mu - |z_i|^2)z_i +$
 442 $\sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation with each cou-
 443 pling weighted by a ratio of amplitudes (`amplitude_phase_is_kuramoto`).
 444 The phase model is this one with the amplitude ratios set to one, rather
 445 than a separate object.

446 What the added degree of freedom decides is which windings exist at all.
 447 On a periodic array of any dimension, this sheet included, a winding requires
 448 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,

449 summed over separations, with ϕ_d the phase the winding advances across d
 450 (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
 451 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
 452 lowers it further. A broad enough kernel therefore removes the winding out-
 453 right. The sweep above reaches that same boundary by a different route:
 454 the phase model retains every winding at every coupling, and loses this one
 455 under integration instead. The two layers bound the same regime from op-
 456 posite sides. A phase recording does not separate a state the amplitude layer
 457 excludes from one the dynamics fails to sustain.

458 These states are windings and not spirals, and on the sheet that is a
 459 theorem rather than a shortage of dimensions. A cortical phase singularity
 460 is a point at which the amplitude vanishes and around which the phase turns
 461 once, read in practice as the circulation around a small plaquette. The states
 462 above have equal modulus at every site, so that circulation is exactly zero on
 463 every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can the
 464 layer host a single defect. Over a periodic sheet the circulations of *any* phase
 465 field sum to zero, so defects arrive in cancelling pairs (`no_isolated_defect`).
 466 The sweep's near-zero measured defect density agrees with the first of these
 467 and says nothing beyond it.

468 Two limits bound the layer for a reader of the cortical literature. Relax-
 469 ational dynamics has no propagation speed, which is the quantity a cortical
 470 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
 471 sured envelope is not; this work identifies the two nowhere. Supplemental

472 Material gives the sweep, its uniform control, the amplitude layer and the
 473 standing wave it admits inside the band, together with the remaining limits
 474 of both.

475 A vanishing resultant is the absence of global agreement and not of
 476 agreement. Eq. (6) read on a nearest-neighbour patch of a winding gives
 477 a discrepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the
 478 patch, which falls to zero as the same winding is spread over more sites
 479 (`chord_le_of_char_patch`), while the global form of the same bound is vac-
 480 uous at $r = 0$. Encoders shared on overlaps therefore still bound content
 481 disagreement inside a patch of a state that no global observable calls coher-
 482 ent. The winding number itself stays unconnected to content.

483 This qualifies what a phase trace can establish. An onset criterion stated
 484 in r alone registers a change of basin as readily as a change of coupling.
 485 The protocol of Section 7.3 therefore needs a spatial observable beside the
 486 resultant.

487 **6.4 Coherent adaptation can lose environmental align-** 488 **ment**

489 An adaptive coupling rule might supply representational structure, but its
 490 objective must reward the relevant agreement. For symmetric coupling, pair-
 491 wise sine terms cancel in the population mean drift. Minimizing squared drift
 492 at that fixed mean therefore rewards equalizing drift across nodes. In the

493 frequency-cluster control, this favors coupling between unlike-frequency clus-
494 ters, whereas the diagnostic environmental template rewards coupling within
495 clusters.

496 The formalization proves descent of the squared-drift objective for a spec-
497 ified gradient flow at fixed phases. Its functional differs from thermodynamic
498 entropy production: at a coherent stationary density the probability current
499 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
500 rial gives the operator, its gradient and the conditions of the descent result.

501 The finite control co-evolves noisy phases and symmetric, nonnegative
502 couplings at fixed total resource. Its template is a diagnostic covariance of
503 latent signals, never a training input. Across three seeds, second-half order
504 remains between 0.9167 and 0.9222, while the within-cluster to between-
505 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
506 Random-update, relabelled-gradient and frozen-kernel controls are reported
507 in Supplemental Material.

508 The immediate post-update objective reduction disappears when aver-
509 aged over complete update intervals. A prespecified grid of 6 rate–cadence
510 combinations contains no case meeting the 5% sustained-reduction criterion
511 while retaining coherence and losing alignment. The finding is coherent adap-
512 tation away from the specified structure, with no demonstrated sustained cost
513 reduction.

514 Kernel alignment and overlap compatibility are different observables.
515 This control addresses acquisition of its designated template under its par-

516 ticular objective. Learning models can explicitly include data in an objective
 517 (Corberi et al., 2026); an account of agreement about the cup must specify
 518 the content its learning rule is meant to preserve.

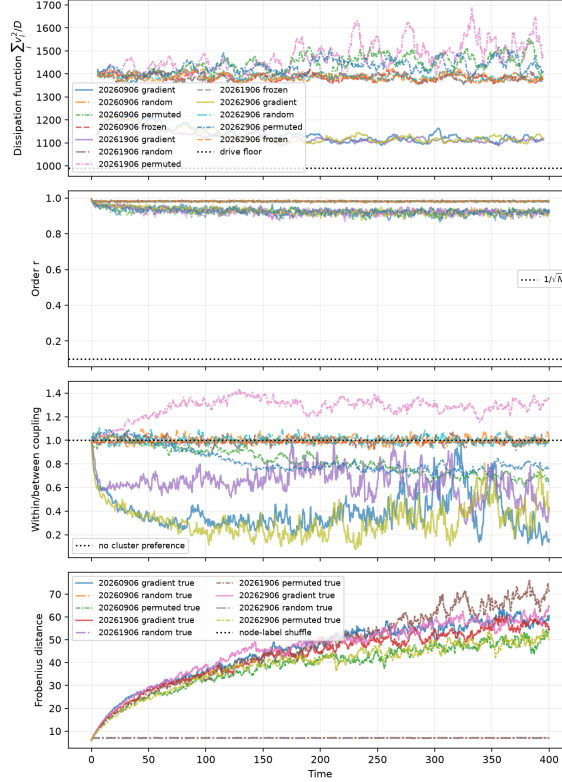


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

519 **6.5 Content agreement needs independent decoders**

520 A neural test needs independently identified domains, local descriptions and
521 restriction maps. Two decoders should estimate a shared content on over-
522 lapping domains before their agreement is examined. Discrepancy requires
523 a null under decoding noise and controls using deliberately compatible and
524 incompatible examples. An implementation on constructed data separates
525 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
526 trial; the compatible construction scores 0.476 on an interval containing one
527 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
528 regional bias can manufacture disagreement. Per-patch decoding diagnostics
529 detect these failures; the compatibility statistic alone does not.

530 Phase agreement cannot replace that measurement. On constructed con-
531 tent, phase carries a share 1.000 of the content entropy when content is built
532 from it and 0.000 when it is not. A phase-derived statistic reports com-
533 patibility on phase-locked patches whose local descriptions disagree. The
534 implemented content statistic detects their incompatibility. These controls
535 validate a measurement on known constructions; a neural content model and
536 decoder remain to be established.

537 Choosing the cover also chooses a candidate subject: trivial descriptions
538 can agree while carrying little content, and a partition can conceal disagree-
539 ment. The useful choice must therefore be constrained by what neural ac-
540 tivity encodes. The exact composition assumes compatibility of its supplied
541 cover. The content mechanisms above identify additional hypotheses under

542 which agreement can arise; they leave the empirical choice of content and
543 the connection to that cover explicit.

544 **6.6 Phase-density estimation and measurement feasi-** 545 **bility**

546 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
547 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
548 itages produce different residuals. The bipolar results are compatible with the
549 Bessel relation, but their concentration range cannot distinguish it from its
550 linear approximation at the observed scatter. Temporal pooling changes the
551 estimand, and sedation recordings leave the awakening hypothesis untested.

552 The concentration range and dependence determine whether the relation
553 can be tested. Under the specified log-density estimator and bias-floor crite-
554 rion, discrimination at the observed concentration ceiling $a = 0.542$ requires
555 3000 independent sites in the scanned designs, rising to 31000 when sites
556 share a mean in groups of 10. These counts describe that estimator and de-
557 pendence model; the intended recording requires its own calibration. Supple-
558 mental Material reports the bin-count controls and the full dependence-aware
559 design.

560 An informative dataset needs sufficient concentration range or precision,
561 calibrated estimators and an observation model for dependence, source mix-
562 ing and reference choice. Public mouse calcium imaging provides thousands

563 of spatially located neurons across waking, sleep and anesthesia (Oomoto et
564 al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used
565 here. Site count, state coverage and bandwidth are separate requirements.

566 **7 The cortical hypothesis and an awakening** 567 **test**

568 The cortical proposal identifies a physical source of coupling and asks whether
569 its changes help explain recovery. The tests in Section 6 determine what
570 observations must distinguish before a phase trace can support this identifi-
571 cation.

572 **7.1 Physical identification and calibration**

573 The cortical hypothesis identifies part of effective coupling with endogenous
574 extracellular electromagnetic interactions. Fields can modulate spike timing
575 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
576 propose coordination of distributed neural information (Pinotsis & Miller,
577 2022; McFadden, 2020). Phase organization under propofol motivates study-
578 ing this regime (Bastos et al., 2021), but does not measure coupling.

579 The formal field predicate requires a jointly continuous kernel, a normal-
580 ized substrate, coupling equal to the kernel average, and no single site pro-
581 viding more than half that strength. Toy constructions satisfy it. A cortical

582 realization must specify the scale at which the kernel and its reduction ap-
 583 ply: calcium imaging reports different state-dependent modularity patterns
 584 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

585 The spatial reduction is also testable. A row-sum aggregation at zero
 586 diagonal tracks the uniform-coupling control over the tested spatial decay
 587 range, while frequency heterogeneity displaces the operational threshold. The
 588 fit window biases both estimates relative to $2D$; agreement with the control
 589 is therefore the supported conclusion. Supplemental Material reports the
 590 reduction, its numerical error and the distinct assumptions needed for the
 591 kernel-average identification in Eq. (5).

592 The field commitment requires physical evidence beyond a choice of math-
 593 ematical representation. A finite matrix spread over positive-mass cells pre-
 594 serves its total coupling and phase correlation exactly. Point-supported
 595 vanishing under an atomless measure and a disadvantage from excluding
 596 a better-correlated pair concern specified support constraints. Neither com-
 597 parison supplies energy prices or a biological–digital distinction.

598 A spike-time shift alone supplies no coupling rate. The dimensionless
 599 product of population size N , timing sensitivity s , field amplitude E and
 600 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

601 This rate describes how induced timing shifts accumulate into drift. Nei-

ther it nor the installation factor κ follows from a heat bound. Testing the physical threshold requires independent estimates of the interaction kernel, its aggregation and diffusion in the same cortical regime.

7.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (15) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

This follows from Eq. (3) at fixed D and concerns the stationary branch

621 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
622 dently justified tracking approximation. The crossing controls in Section 6.2
623 leave that approximation unsupported at onset, even though a separation of
624 geometry and phase time scales can support it away from threshold. The
625 awakening hypothesis therefore needs a calibrated dynamical prediction for
626 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
627 does not supply a recovery latency.

628 **7.3 A protocol for the awakening experiment**

629 The physical test has six requirements:

- 630 1. **Coupling and diffusion in the same units.** Estimate them from re-
631 solved phase dynamics, separating heterogeneity and observation noise.
632 Measure the drive or perturb the system to distinguish endogenous cou-
633 pling.
- 634 2. **Geometry calibrated to coupling.** Validate the geometry observ-
635 able and its relation to effective coupling, with uncertainty. The con-
636 ductivity argument supplies only the expected direction.
- 637 3. **Calibrated phase measurements.** Estimate r by resultant length
638 and a by unweighted regression of log density on centered cosine.
639 Choose site count and binning from a dependence-aware calibration
640 over the intended concentration range. The tested designs require

641 thousands of sites at the observed concentrations; temporal pooling
642 changes the spatial estimand.

643 4. **An independent recovery endpoint.** Measure behavioral recovery
644 separately from phase organization.

645 5. **A predeclared dynamical onset criterion.** Set a coherence thresh-
646 old above the finite-size floor and evaluate its latency under the mea-
647 sured coupling trajectory, including ramp speed or fluctuations. Record
648 a spatial observable beside the resultant, since a winding field reports
649 a low resultant at coupling that sustains local order (Section 6.3).

650 6. **Matched model comparison.** Compare coupling increase, diffusion
651 change, common drive and alternative recovery models under the same
652 observation and filtering assumptions.

653 Failure of the concentration–coherence relation over an informative range
654 constrains the density model. Failure of the calibrated geometry–coupling
655 association constrains the physical gate. Better prediction of independent
656 recovery by a competing mechanism counts against its proposed role. A
657 favorable outcome still leaves the content of coordinated activity to be ex-
658 plained.

659 8 Discussion

660 Stated positively, and granting the premises collected in Table 1, the account
661 runs as follows. Extracellular contraction on awakening raises endogenous
662 field amplitude and with it the effective coupling between cortical oscillators.
663 Once that coupling passes $2D$, a coherent phase branch exists, and from an
664 initial spread within a quarter turn the population settles onto it; other initial
665 data leave windings stationary at the same coupling. Encoders shared where
666 cortical territories overlap turn that phase order into agreement about the
667 quantities the territories describe in common, so the local descriptions glue:
668 there is one state of which each is a restriction. Part of that state encodes the
669 state itself, and a readout of that part reconstructs it accurately enough, and
670 sensitively enough across the states that matter, to make it the unique state
671 the system models itself as being. On this account the glued state is what
672 makes distributed activity one situation, and its accurate self-reconstruction
673 is what gives that situation a perspective.

674 Every step of that description is conditional, and its conditions are the
675 subject of what follows.

676 8.1 Unity and a minimal self as empirical commit- 677 ments

678 We propose the state glued from compatible local descriptions as a corre-
679 late of conscious unity, and its accurate internal reconstruction as a correlate

680 of a minimal self. The motivations are that different contents belong to
681 one experienced situation and that the situation is experienced from a per-
682 spective. The conditional results identify relations a realization would need:
683 sufficient installed resources for the specified phase model, agreement about
684 shared content, and an encoding that supports reconstruction at the required
685 fidelity.

686 Compatibility allows different descriptions to contribute different infor-
687 mation. It also permits uncertainty and conflicting judgments: apparent cup
688 position and judged cup position can be distinct quantities in a consistent
689 description of their discrepancy. Which quantities are shared must be de-
690 termined independently. Choosing them solely to secure agreement would
691 deprive the proposal of empirical force.

692 The minimal-self proposal concerns the perceiver's position and condition
693 within the situation, such as the cup's location relative to a body that can
694 reach it. The encoding could be implicit and need carry no autobiograph-
695 ical narrative. Its proposed necessity is philosophical: even an unreflective
696 episode has a perspective. Gluing by itself requires no self-encoding, and re-
697 constructability alone supplies no functional role. Altering the encoded hand
698 or viewpoint while holding the external scene fixed should change estimates
699 of the perceiver's relation to the cup as the readout predicts. Whether that
700 represented perspective is experienced remains a further question.

701 8.2 What a test of the full proposal must establish

702 Both uniqueness results concern specified objects. They choose neither the
703 cover nor the encoding, and do not select between candidate subjects. In-
704 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
705 here selection requires independent evidence about neural organization. A
706 constant readout can contract and fix a predetermined state while discard-
707 ing its input. The family criterion in Section 4.2 tests the sensitivity that
708 this construction lacks, provided the relevant states and fidelity are declared
709 independently.

710 A test must separately measure coupling relative to diffusion, content
711 compatibility and self-reconstruction, alongside an independent endpoint for
712 experience. Meeting all three without a unified episode would challenge suffi-
713 ciency. A unified episode with incompatible descriptions would challenge the
714 unity identification; one lacking the specified self-encoding would challenge
715 the reflexive requirement. Awakening, dreaming and dissociative states offer
716 possible settings, but absence of report cannot by itself define absence of
717 experience.

718 The next physical requirement is a calibrated geometry–coupling relation
719 and resolved phase dynamics. The installation bound contributes a nec-
720 essary resource condition where its hardware representation applies; cortical
721 mode prices and field-mediated coupling require measurement. For the repre-
722 sentational connection, an informative experiment asks whether a controlled
723 change in coordination alters independently decoded content agreement, with

724 controls for sensory information and decoder quality. The constructed-data
725 estimator establishes feasibility for specified contents. Neural local descrip-
726 tions, shared overlap encoders and the self-encoding have yet to be identified.
727 The population-size dependence of the phase-to-content bound further limits
728 its quantitative use.

729 The separation results explain why these measurements must be inde-
730 pendent. A stationary distribution leaves several recovery mechanisms indis-
731 tinguishable. Equal phase order can accompany incompatible descriptions
732 or different task information. The tested plasticity rule preserves coherence
733 while losing its designated template. Each finding concerns the specified
734 model and observable; their role is to delimit the evidence needed for the
735 proposed connections.

736 **8.3 GPUs and the limits of no-go claims**

737 For a large language model (LLM) running on graphics processing units
738 (GPUs), the framework requires identifying a physical system whose local
739 descriptions form one state and whose internal encoding reconstructs that
740 same state. The coupling-gated route E56–E78 requires a calibrated physi-
741 cal coupling, its noise regime and a connection from coordination to content
742 agreement. An attention matrix or a shared clock supplies none of these iden-
743 tifications on its own. Computing a field model likewise leaves its realization
744 in the computing substrate to be established. These physical obligations also
745 apply to the cortical proposal.

746 Unity requires the candidate’s local descriptions to agree on indepen-
747 dently identified shared quantities (Section 3). Common timing and fluent
748 output do not establish that agreement. Digital representations can sat-
749 isfy compatibility, however, so finite hardware is compatible with the gluing
750 construction. The empirical commitment is to identify the relevant cover
751 and contents with one experienced situation; mathematical uniqueness alone
752 selects no subject.

753 For self, the system must carry an encoding $E(s)$ and an internal readout
754 R of the same global state. E89 requires $F = R \circ E$ to contract and fix
755 the state produced by gluing. A constant readout shows why a unique fixed
756 point alone is insufficient for informative self-representation. The additional
757 family criterion in Section 4.2 supplies a test: if two relevant macrostates
758 satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both
759 within error ε . An intervention outside the encoding region’s causal past
760 gives a corresponding obstruction before a declared communication deadline.
761 Identical verbal reports alone do not establish identical internal codes. These
762 conditional no-go results exclude a specified reconstruction at a specified
763 fidelity or deadline. They apply to biological and digital candidates alike.

764 The hardware comparisons in Supplemental Material have similarly re-
765 stricted scope. Fixed wiring can impose a correlation disadvantage, and
766 point-supported kernels vanish under an atomless continuum measure. Nei-
767 ther comparison makes the representation conditions unattainable on GPUs.
768 A universal exclusion would need a necessary condition for consciousness and

769 a proof that every relevant implementation violates it. The present condi-
770 tional composition supplies neither.

771 Whether experience requires a living substrate is the explicit subject of
772 biological naturalism (Seth, 2026a,b). On that view the properties ground-
773 ing experience are those of self-sustaining organisms, and simulating them
774 does not secure them. The conditions stated here name no biological mecha-
775 nism. Computation described as such does not satisfy them either: a candi-
776 date must exhibit a physically realized coupling in a measured noise regime,
777 agreement about independently identified shared quantities, and an internal
778 reconstruction meeting a declared fidelity. Whether a realization of those
779 conditions that is not alive would be accompanied by experience is the point
780 at issue, and these results do not settle it. A related position locates the
781 difference in the mode of computation rather than in life as such: biologi-
782 cal systems compute across scales that do not separate, and alongside dis-
783 crete operations carry continuous-valued dynamics belonging to their phys-
784 ical medium (Milinkovic & Aru, 2026). The coupling condition stated here
785 is one measurable instance of that requirement, fixed to a physical field in a
786 declared noise regime, and a description of such a field does not satisfy it.

787 The neuroscientific case against *human-like* consciousness in text-only
788 LLMs concerns embodied sensory interaction, thalamocortical organization
789 and biological regulation linking perception and action to survival (Aru et
790 al., 2023). Their absence supports skepticism, conditional on their causal
791 relevance to experience. Those mechanisms are not requirements of the ab-

792 stract gluing and reconstruction definitions. Workspace-like representations
793 reported in some LLMs (Gurnee et al., 2026) provide candidate organization
794 to investigate. Within this framework, the relevant evidence would identify
795 compatible contents and an informative internal reconstruction, or demon-
796 strate their failure under controlled interventions.

797 **8.4 Explanatory scope**

798 Integrated-information, global-workspace and predictive-processing accounts
799 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
800 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
801 candidate physical interaction and conditional relations between local content
802 and self-encoding. Gluing and reconstruction permit multiple substrates;
803 the electromagnetic commitment enters through physical calibration. The
804 formalization checks implications under their hypotheses, and its witnesses
805 establish realizability on declared models. They do not together supply a
806 cortical agent.

807 The explanatory question is whether the specified relations occur in neural
808 activity and matter for experience. In the cup example, this means establish-
809 ing how coordination helps maintain one represented situation and how the
810 perceiver’s place within it contributes to the episode. The proposal identi-
811 fies candidate correlates and tests of that identification. Why those relations
812 would be accompanied by experience remains unexplained.

813 **Data and code availability**

814 The Lean development, simulation scripts, tests and saved numerical sum-
815 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
816 `cortical_coherence`. Publication simulation values are generated from
817 saved summaries without rerunning production sweeps. The exploratory
818 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
819 processing and selection are described in Supplemental Material. The repos-
820 itory contains the reproducible analysis and does not redistribute the raw
821 recordings.

822 **Funding and competing interests**

823 This research received no external funding. The author declares no compet-
824 ing interests.

825 **Use of AI tools**

826 AI assistance to this work was substantial and is not separable by task. Sev-
827 eral coding assistants and chat interfaces were used in an interleaved way
828 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
829 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
830 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
831 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,

832 accessed with the Hermes CLI. Between them they contributed to the Lean
833 development, the simulation and analysis code, and the drafting and revision
834 of both documents. Because the tools were used in combination and in alter-
835 nation, no section, proof, figure or result is attributable to any one of them,
836 and none is claimed to be. The author is responsible for all content, including
837 everything a tool produced. Machine checking establishes the stated Lean
838 results under their hypotheses; it does not validate biological interpretations
839 or replace scientific review of generated code and prose.

840 References

- 841 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
842 electric fields emerge from and allow representational drift. *NeuroImage*,
843 253, 119058.
- 844 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
845 tion theory: from consciousness to its physical substrate. *Nature Reviews*
846 *Neuroscience*, 17(7), 450-461.
- 847 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
848 Mind. *Oxford University Press*.
- 849 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
850 tational theory of conscious processing. *Current Opinion in Neurobiology*,
851 25, 76-84.

- 852 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
853 *Reviews Neuroscience*, 11(2), 127-138.
- 854 Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness:
855 A case for biological computationalism. *Neuroscience & Biobehavioral Re-*
856 *views*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.
- 857 Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial
858 consciousness through the lens of neuroscience. *Trends in Neurosciences*,
859 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.
- 860 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
861 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- 862 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
863 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
864 doi:10.1017/S0140525X26106906.
- 865 Gurnee, W., et al. (2026). Verbalizable Representations Form a
866 Global Workspace in Language Models. *Transformer Circuits Thread*.
867 arXiv:2607.15495.
- 868 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
869 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
870 40, 24-50.
- 871 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
872 parts of the free energy principle. *Entropy*, 23(3), 293.

873 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
874 peror's New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

875 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
876 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

877 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
878 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
879 *Progress in Neurobiology*, 226, 102465.

880 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
881 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
882 9(8), 45-60.

883 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
884 testable hypothesis about the characteristics of conscious as opposed to
885 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

886 McFadden, J. (2020). Integrating information in the brain's EM field: the
887 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
888 niaa016.

889 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
890 guide neocortical network activity. *Neuron*, 67(1), 129-143.

891 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
892 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
893 95-103.

- 894 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
895 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 896 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
897 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 898 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
899 *logical Reviews*, 98(1), 239-389.
- 900 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
901 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
902 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 903 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
904 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
905 ity of complexity measures to capture dreaming during propofol sedation.
906 *Scientific Reports*, 15, 32746.
- 907 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
908 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
909 brain. *Science*, 342(6156), 373-377.
- 910 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
911 *Physiological Reviews*, 88(4), 1277-1340.
- 912 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
913 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

914 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
915 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
916 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

917 Landauer, R. (1961). Irreversibility and heat generation in the computing
918 process. *IBM Journal of Research and Development*, 5(3), 183-191.

919 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
920 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

921 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
922 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

923 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
924 of prediction. *Physical Review Letters*, 109(12), 120604.

925 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
926 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

927 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
928 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
929 031015.

930 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
931 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
932 PhysRevLett.111.180603.

- 933 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
934 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 935 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
936 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
937 181.
- 938 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
939 archive.pitt.edu/115.
- 940 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
941 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
942 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 943 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
944 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
945 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 946 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
947 bifurcations with additive noise. *Probability Theory and Related Fields*,
948 122, 341–388. doi:10.1007/s004400100174.
- 949 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
950 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
951 doi:10.1088/1367-2630/13/11/113036.
- 952 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
953 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave

- 954 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
955 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- 956 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
957 terns underlie complex brain dynamics and are related to cognitive pro-
958 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
959 023-01626-5.
- 960 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
961 and dissipation with partially observed fluctuating signals. *Physical Review*
962 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.
- 963 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
964 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
965 cell resolution functional networks during unconsciousness are segre-
966 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
967 doi:10.1016/j.celrep.2025.116902.
- 968 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
969 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
970 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.